

Measuring AI Ability to Complete Long Software Tasks

50% TASK-COMPLETION TIME HORIZON DOUBLING EVERY ~7 MONTHS

GPT-2 completed 2-second tasks (2019) → o3 now handles ~2-hour tasks (2025)

Thomas Kwa, Ben West, Joel Becker et al. — arxiv.org/abs/2503.14499

FOR: AI Safety Researchers · Frontier AI Lab Leadership · Policy Analysts

A New Metric: 50% Task-Completion Time Horizon

Three-step methodology produces a human-interpretable measure of autonomous AI capability

1

Diverse Task Suite

Assemble 170 software & research tasks spanning 1 second to 8 hours of skilled human effort

2

Human Timing + AI Testing

Time 800+ human baselines (2,529 hrs total). Run each AI agent on tasks and record pass/fail results

3

Logistic Model Fit

Fit logistic curve to AI success rates vs. task duration. Read off the 50% success point as the "time horizon"

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Figure 2 — Methodology overview (METR, 2025)

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST

97 tasks · 1 min – 30 hrs

RE-Bench

7 tasks · 8 hrs (all)

SWAA

66 tasks · 1 sec – 30 sec

800+ human baselines collected · 2,529 hours of expert labor

Task Name	Suite	Human Time	Description / Example
find_shell_script	SWAA	3 seconds	Multiple choice: 'Which file is a shell script?' — very short single-step task
wikipedia_research	SWAA	1 minute	Research simple factual information from Wikipedia
oxdna_simple	HCAST	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	HCAST	56 minutes	Write Python script to transform JSON data by inferring conversion rules
cuda_backtesting	RE-Bench	8 hours	Implement custom CUDA kernels for 30× speedup of a backtesting tool

Task durations reflect skilled human professional completion times. HCAST = Human-Computer Autonomy Software Tasks.

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hr 50 Min (o3) in 6 Years

~207 days

Doubling Time

$R^2 = 0.97$

Exponential Fit (6 yrs)

2 sec → ~110 min

GPT-2 to o3

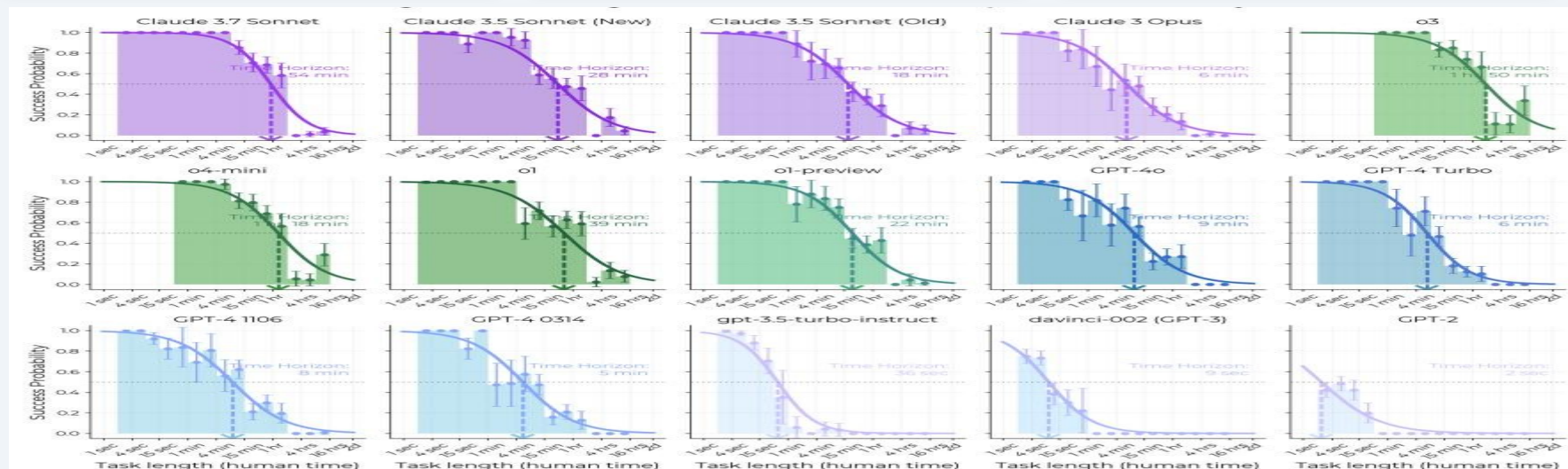
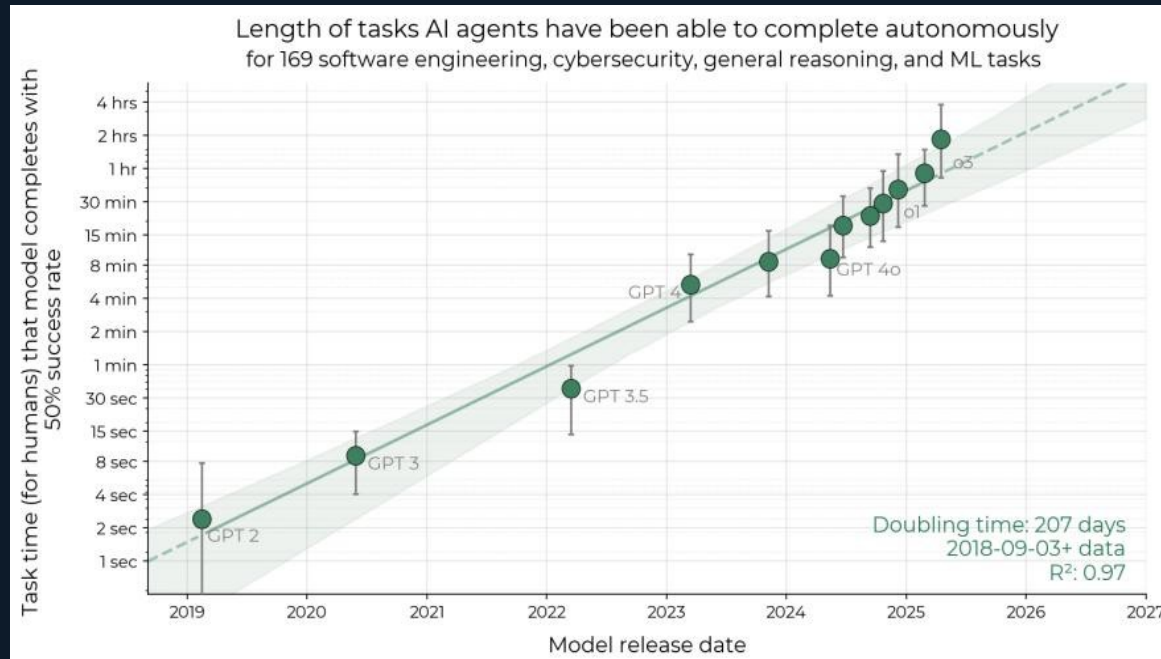


Figure 1 — 50% task-completion time horizon vs. model release date (METR, 2025). Log scale. Doubling time ≈ 207 days, $R^2 = 0.97$.

Success Probability Drops as Task Length Increases — Frontier Models Push the Curve Right



50% TIME HORIZONS

GPT-2

2019 · ~2 sec

GPT-4 (0314)

2023 · ~5 min

Claude 3.5 Sonnet

2024 · ~28 min

o1

2024 · ~39 min

Claude 3.7 Sonnet

2025 · ~54 min

o3

2025 · ~1 hr 50 min

Figure 3 — Per-model success probability vs. task duration (METR, 2025).

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Improved Logical Reasoning

Models reason through multi-step plans, debug systematically, and handle complex dependencies across long task sequences.

Better Tool Use

Improved ability to call shell commands, write/run code, read files, and interact with APIs with greater accuracy and fewer failures.

Reliability & Self-Awareness

Frontier models more consistently recognize when they are off-track and course-correct before failures compound.

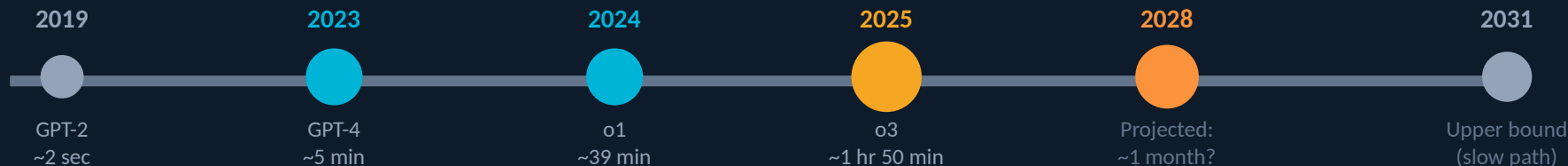
Ability to Adapt to Mistakes

When errors occur, models increasingly recover by re-trying, re-planning, or escalating — rather than silently failing.

LIMITATION:

Performance is notably lower on less structured, 'messier' tasks. External validity is an open question — the benchmark may not perfectly represent the average segment of real-world intellectual labor.

Extrapolation Predicts Month-Long Task Automation by 2028–2031



◆ ~20% faster growth in 2023–2025 vs. the full 2019–2025 trend suggests possible acceleration.

◆ Naive extrapolation: 1-month time horizon (167 work hours) between mid-2028 and mid-2031.

◆ 80% time horizon is ~5× shorter than the 50% horizon — tasks at the capability frontier remain hard.

Caveats: Trend assumes no inflection. Benchmark tasks may not match real-world job distributions. External validity unverified.

Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1

Robust & Interpretable Metric

50% time horizon is a human-grounded, statistically robust metric for AI autonomy — $R^2=0.97$ over 6 years across 10+ models.

2

Doubling Every ~7 Months

The trend has been remarkably consistent since 2019, with possible acceleration since 2024 (~20% faster in 2023–2025 vs. full period).

3

Frontier Already at ~2 Hours

o3 (2025) can autonomously complete tasks taking skilled humans ~1 hr 50 min — up from 2 seconds for GPT-2 in 2019.

4

2028–2031: Month-Long Automation

Extrapolation suggests AI reaches a 1-month (167 hr) time horizon between mid-2028 and mid-2031, enabling automation of many software projects.

5

Limitations & Open Questions

Performance degrades on unstructured tasks; external validity to real-world job distributions remains unverified. Trend extrapolations assume no structural changes.